finite dimensional Lie algebra $\mathfrak{g}$ over a field k is said to be **solvable** if there exists n such that $K^n(\mathfrak{g}) = 0$. Prove the Lie theorem: if $k = \mathbb{C}$ and V is a finite dimensional irreducible representation of a solvable Lie algebra $\mathfrak{g}$, then V is 1-dimensional.

Hint: Prove the result by induction in dimension. By the induction assumption, $K(\mathfrak{g})$ has a common eigenvector v in V; that is, there is a linear function $\chi : K(\mathfrak{g}) \to \mathbb{C}$ such that $av = \chi(a)v$ for any $a \in K(\mathfrak{g})$. Show that $\mathfrak{g}$ preserves common eigenspaces of $K(\mathfrak{g})$. (For this you will need to show that $\chi([x, a]) = 0$ for $x \in \mathfrak{g}$ and $a \in K(\mathfrak{g})$. To prove this, consider the smallest subspace U containing v and invariant under x. This subspace is invariant under $K(\mathfrak{g})$ and any $a \in K(\mathfrak{g})$ acts with trace $\dim(U)\chi(a)$ in this subspace. In particular $0 = \mathrm{Tr}([x, a]) = \dim(U)\chi([x, a])$.)

Problem 2.16.2. Classify irreducible finite dimensional representations of the two-dimensional Lie algebra with basis X, Y and commutation relation $[X, Y] = Y$. Consider the cases of zero and positive characteristic. Is the Lie theorem true in positive characteristic?

Problem 2.16.3. (Hard!) For any element x of a Lie algebra $\mathfrak{g}$ let $\mathrm{ad}(x)$ denote the operator $\mathfrak{g} \to \mathfrak{g}, y \mapsto [x, y]$. Consider the Lie algebra $\mathfrak{g}_n$ generated by two elements x, y with the defining relations $\mathrm{ad}(x)^2(y) = \mathrm{ad}(y)^{n+1}(x) = 0$.

(a) Show that the Lie algebras $\mathfrak{g}_1, \mathfrak{g}_2, \mathfrak{g}_3$ are finite dimensional and find their dimensions.

(b) (Harder!) Show that the Lie algebra $\mathfrak{g}_4$ has infinite dimension. Construct explicitly a basis of this algebra.

Problem 2.16.4. Classify irreducible representations of the Lie algebra $\mathfrak{sl}(2)$ over an algebraically closed field k of characteristic $p > 2$.

Problem 2.16.5. Let k be an algebraically closed field of characteristic zero, and let $q \in k^\times, q \neq \pm 1$. The quantum enveloping algebra $\mathcal{U}_q(\mathfrak{sl}(2))$ is the algebra generated by e, f, K, K^{-1} with relations

$$KeK^{-1} = q^2 e,\ KfK^{-1} = q^{-2} f,\ [e, f] = \frac{K - K^{-1}}{q - q^{-1}}$$

(if you formally set $K = q^h$, you'll see that this algebra, in an appropriate sense, "degenerates" to $\mathcal{U}(\mathfrak{sl}(2))$ as $q \to 1$). Classify irreducible